Problem Set 2 Expectation, Consumption, Investment(Solution)

1. What is the impact on current consumption of a temporary tax cut according to:

(a) the Keynesian consumption function?

(b) the permanent-income hypothesis?

(a) According to the Keynesian consumption function, the current real consumption is determined by the current real income. When the government implements temporary tax cuts, it increases disposable income in the current period, leading to a significant increase in real consumption in the current period.

(b) According to the permanent income hypothesis, the current consumption is determined by permanent income. When the government adopts temporary tax cuts, it only results in a slight increase or even negligible increase in permanent income. Therefore, under the assumption of permanent income, the current consumption will only experience a marginal increase or even almost no increase.

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(a) 依據 Keynesian consumption function 可知, 當期實質消費水準決定於當期實質所得水準。當政府執行暫時性的減稅政策, 會使當期可支配所得增加, 導致當期實質消費顯著的增加。

(b) 依據 permanent-income-hypothesis, 2; 當期消費水準決定於恆常所得。當政府採取暫時性的減稅政策時, 只會使恆常所得微幅增加甚至幾乎沒有增加, 因此在恆常所得假說下, 當期消費只會微幅增加甚至幾乎沒有增加。

2. A firm is considering the purchase of a capital good that will generate an additional \$400 income each year for 4 years(after which time the capital good is useless and has no scrap value). The capital good will cost \$1,600. If the interest rate is 3 percent, what actions should the firm take?

$$NPV = \frac{400}{(1+3\%)} + \frac{400}{(1+3\%)^2} + \frac{400}{(1+3\%)^3} + \frac{400}{(1+3\%)^4} - 1,600 = -113.16 < 0$$

$$NPV = \frac{400}{1.03} + \frac{400}{1.03^2} + \frac{400}{1.03^2} + \frac{400}{1.03^4} - 1,600 < 0$$

$$-100 + \frac{500}{1.02} + \frac{500}{1.02^2} + \frac{500}{1.02^3} =$$

490.2 480.58 471.16

Firm should not purchase the capital good.

3. Consider the payment streams listed below that are available from different capital projects for Shioupi Software. The firm must choose to implement just one out of the three possible projects.

	Today	1 Year	2 Years	3 Years
Retool Engineers' Offices	$-\$100$	\$500	\$500	\$500
Rewire Network	$-\$50$	\$1,000	\$500	\$200
Move to Kaohsiung	$-\$200$	\$300	\$600	\$1,200

(a) If the interest rate were 2%, what actions should Shioupi Software take?

(b) If the interest rate were 20%, what actions should Shioupi Software take?

(a)

$$NPV_1 = -100 + \frac{500}{(1+2\%)} + \frac{500}{(1+2\%)^2} + \frac{500}{(1+2\%)^3} \approx 1,341.94$$

$$NPV_2 = -50 + \frac{1000}{(1+2\%)} + \frac{500}{(1+2\%)^2} + \frac{200}{(1+2\%)^3} \approx 1,599.44$$

$$NPV_3 = -200 + \frac{300}{(1+2\%)} + \frac{600}{(1+2\%)^2} + \frac{1,200}{(1+2\%)^3} \approx 1,801.61$$

Since NPV_3 is the largest NPV, Shioupi Software should choose "Move to Kaohsiung".

4. Suppose an interest rate of 4%, a rate of depreciation of 6%, a price of capital that rises at 1% per year, and a zero corporate tax rate. What is the user cost of capital? Let P_K be the user cost of capital, then

$4 + 6 - 1 = 9\%$

$$P_K = i - \pi + \delta = 4\% - 1\% + 6\% = 9\%$$

text book : rental cost = $r + \delta$

5.If Acer Corporation has current and future marginal productivity of capital given by $MPK = 6,000 - 3K + N$, and marginal productivity of labor given by $MPL = 80 - 1.5N + 2.5K$. The price of capital is $NT\$3,000$, the real interest rate is 2%, and capital depreciation at a 10%. The real wage is $NT\$12$.

$6,000 - 3K + N = 360$

$80 + 2.5K + 1.5N = 12$

$\Rightarrow 3K + N = 5,640$

$-2.5K + 1.5N = 68$

$3{,}000 \times (2\% + 10\%) = 360$

$K = 4{,}264$
$N = 7{,}152$

(a) Calculate the user cost of capital.

(b) Find the Acer's optimal amount of employment and the size of the capital stock.

(a) Let P_K be the user cost of capital.

$$P_K = 3,000(2\% + 10\%) = 360$$

(b) Optimal factor employment decision:

$$MPK = P_K \Rightarrow 6,000 - 3K + N = 360 \rightarrow 3K - N = 5,640$$

$$MPL = w \Rightarrow 80 - 1.5N + 2.5K = 12 \rightarrow -2.5K + 1.5N = 68$$

Combine the two equations above, we can solve

$$K^* = 4,264, \;\; N^* = 7,152$$

6. In terms of the permanent-income hypothesis, would you consume more of your Christmas bonus if

(a) you knew there would be a bonus every year?

(b) this was the only year the bonus would be given?

(a) If you expect to get a Christmas bonus every year from now on, you immediately treat it as part of your permanent income and spend accordingly, that is, $\Delta C = c(\Delta Y_P)$. In other words, your current consumption will change significantly.

(b) If you get a Christmas bonus for only this year but not for any future years, then you will consider it as transitory income. There is very little effect on your permanent income, so you will consume only a small fraction of the bonus and save the rest. In other words, your current consumption will not be significantly affected.

7. What are the similarities between the life-cycle and the permanent-income hypotheses? Do they differ in their approaches to explaining why the long-run MPC is greater than the short-run MPC? Both the life-cycle theory and the permanent-income theory try to explain why the short-run MPC is smaller than the long-run MPC. The life-cycle theory attributes the difference to the fact that people prefer a smooth consumption stream over their lifetime and thus average expected lifetime income is the true determinant of current consumption. The permanent-income theory suggests that the difference is due to measurement errors. Measured income has two components—permanent and transitory income. But only permanent income is a true determinant of current consumption.

8. Suppose that permanent income is calculated as the average of income over the past five years; that is,

$$Y_P = \frac{(Y + Y_{-1} + Y_{-2} + Y_{-3} + Y_{-4})}{5}.$$

Suppose further that consumption is given by $C = 0.9Y_P$.

(a) If you have earned \$20,000 per year for the past 10 years, what is your permanent income? $Y_{P_0} = \frac{1}{5}(20{,}000 \times 5) = 20{,}000$

(b) Suppose that next year (period t + 1) you earn \$30,000. What is your new Y_P?

$Y_{P_1} = \frac{1}{5}(30{,}000 + 4 \times 20{,}000)$
$= 22{,}000$

next year $= 0.9 \times 22{,}000 = 19{,}800$

(c) What is your consumption this year and next year?

this year $= 0.9 \times 20{,}000 = 18{,}000$

(d) What is your short-run marginal propensity to consume? Long-run MPC?

SR: $MPC = 0.9 \times \frac{1}{5} = 0.18$ LR $= 0.9$

(a) If income remains constant over time, then permanent income equals current income. Your permanent income this year is $Y_{P0} = (1/5)(5 \times 20,000) = 20,000$.

(b) Your permanent income next year is $Y_{P1} = (1/5)(30,000 + 4 \times 20,000) = 22,000$.

(c) Since $C = 0.9Y_P$, your consumption this year is $C_0 = 0.9 \times 20,000 = 18,000$. Your consumption next year is $C_1 = 0.9 \times 22,000 = 19,800$.

(d) In the short run, the $MPC = (0.9)(1/5) = 0.18$; but in the long run, the $MPC = 0.9$.